

IC 241: Materials Science for Engineers

Course Code	IC 241
Course Name	Materials Science for Engineers
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech.
Mutual Exclusion	None
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum; 32nd BoA (Dec 2019); 24th Senate (Feb 2020), in the B.Tech-M.Tech Integrated Dual Degree in Bioengineering curriculum

Course Contents

- **Introduction:** Overview of materials science and materials engineering, Materials tetrahedron; structure – property correlations, processing, Materials selection strategies; Property considerations for specific application, Ashby-style charts, Impact of bonding and structure over materials properties. Classes of engineering materials; metals, polymers, ceramics, composites, and their properties. Crystal structures and microstructures in materials: Amorphous, crystalline, and polycrystalline materials, Crystalline defects, and their significance. [10 Hours]
- **Structural Materials and Applications:** Iron, steel, light weight metals, alloys, polymers and composites, materials for automotive, railways, aircrafts, and the future VTOLs and Hyperloops, defense, marine and oil & gas sectors. Uniaxial stress, strain, engineering and true stress and strain, stress strain diagrams, elastic, yielding and plastic behaviour, effect of temperature and creep. Application of metals, ceramics, polymers, and composites in static structures like buildings, bridges, furnace structure, bulb filaments, etc, strength requirement in transmission lines. [10 Hours]
- **Optoelectronic Materials and Applications:** Band structures for conductors, semiconductors and insulators, I-V characteristics, resistance of alloy, and conductor alloy. Light interactions with solids, refraction, reflection, absorption and transmission, luminescence, photoconductivity, lasers and optical fibre communications, photovoltaics, light emitting diodes, photodetector. [6 Hours]
- **Dielectric, Magnetic, Superconductors and their applications:** Dielectric Materials - Polarization mechanisms, frequency dependence of dielectric constant and dielectric loss, piezo, ferro and pyro-electricity-elemental ideas. Choice of materials for various specific applications: capacitors, sensors, actuators, and transducers, in the context of applications. Magnetic materials: dia, para, ferro, antiferro and ferrimagnetism. Soft and Hard magnetic materials, magnetic sensors, read- write heads, spintronic devices. Superconducting materials; zero resistance and Meissner effect. Type I and Type II superconductors. High temperature superconducting materials and their applications. [10 Hours]
- **Advanced Materials and Sustainability:** Nanomaterials, Biomaterials, Smart Materials, Energy Materials, Materials for clean energy and carbon capture, Recycling, Waste management, Environmental Degradation, Sustainability and Green Engineering etc. [6 Hours]

Text Books

1. William D. Callister, Jr., **Materials Science and Engineering- An introduction**, John Wiley and Sons, Inc.
2. James F Shackelford, **Introduction to Materials Science for Engineers**.
3. Lawrence H. Van Vlack, **Elements of Materials Science and Engineering**, Addison-Wesley Publishing Company.
4. W.F. Smith, **Principles of Materials Science and Engineering**, McGraw-Hill.
5. R.A. Flinn and P.K. Trojan, **Engineering Materials and Their Applications**, Haughton.

References

1. Kenneth G. Budinski, **Engineering Materials: Properties and Selection**, Prentice Hall, [New Edition] USA.
2. S. O. Kasap, **Principles of Electronic Materials and devices**.